

NAGARJUN GOWDA KANCHANAHALLI NINGE GOWDA

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EDUCATION

BMS Institute of Technology and Management, Bengaluru

2023 – 2027 (expected)

B.E. Computer Science & Engineering CGPA: 9.38/10.0 Top 5% of a class of about 500 Class XII: 95.5% Class X: 94.4%

Relevant Coursework: Data Structures & Algorithms, Operating Systems, Database Systems, Computer Networks, Machine Learning

PUBLICATIONS

Bold = applicant. All venues are peer-reviewed.

- [1] **Nagarjun Gowda K N** and Dhanalakshmi B K, "AI-NIDPS: A Deployment-Oriented Flow-Centric Architecture for Real-Time Explainable Network Intrusion Detection," in *Proc. 2nd Int. Conf. Computing for Sustainability and Intelligent Future (COMP-SIF)*, 2026, doi: 10.1109/COMP-SIF69752.2026.11481940. *Published, IEEE Xplore.*
- [2] **Nagarjun Gowda K N** and Dhanalakshmi B K, "Energy-Aware Adaptive Inference for Computationally Efficient Deep Learning," in *Proc. 2nd Int. Conf. Next Generation Electronics (NEleX)*, 2026, doi: 10.1109/NELeX69209.2026.11590454. *Published, IEEE Xplore.*
- [3] **Nagarjun Gowda K N**, Jayaraj U B, and Muthukumaran L, "Scale-Invariant Sharpness Measures and Their Impact on Generalization in Deep Neural Networks," *IEEE I4C 2026*. *Accepted and presented.*
- [4] **Nagarjun Gowda K N**, Dhanalakshmi B K, and Anusha K L, "AgentShield: A Cooperative Multi-Agent Reinforcement Learning Framework for Autonomous Detection and Response to Novel Network Intrusions," *IEEE International Flagship Conference, NKCon-2K26*. *Accepted.*
- [5] **Nagarjun Gowda K N** and Ashwini N, "Clinical and Linguistic Metrics Converge on Distinct Timescales: A Lightweight Dual-Adapter Framework for Chest X-Ray Report Generation," *IEEE International Flagship Conference, NKCon-2K26*. *Accepted.*
- [6] **Nagarjun Gowda K N** and Dhanalakshmi B K, "Topology-Aware Explainability with Conformal Uncertainty Quantification for Real-Time Payment Fraud Detection: A Graph-Augmented Streaming Framework," *Expert Systems with Applications (Elsevier)*. *Under review (final stage).*
- [7] **Nagarjun Gowda K N** and Ashwini N, "Coverage-Degradation Bounds for Conformal Prediction Under Energy-Aware Classifier Switching: A Case Study on Network Intrusion Detection Benchmarks," *IEEE Access*. *Under review.*
- [8] **Nagarjun Gowda K N** and Yashwanth K R, "CausalStress: Counterfactual Stress Testing and Causal-Consistency Evaluation for Multivariate Time-Series Forecasters," *4th Int. Conf. Data Science and Information System (ICDSIS-2027)*. *Submitted.*

RESEARCH EXPERIENCE

AgentShield: Autonomous Cyber Defense with Multi-Agent RL

NSL-KDD, UNSW-NB15

- **Problem/Action:** ML intrusion detectors stop at classification, so I built a five-agent cooperative RL system (detection, analysis, containment, recovery, coordinator) whose conformal risk calibration drives graduated responses (log $\rightarrow$ rate-limit $\rightarrow$ quarantine $\rightarrow$ isolate).
- **Result:** Diagnosed a failure of fixed conformal calibration (zero TPR on held-out NSL-KDD U2R, Probe, R2L) and designed a saturation-gated threshold that recovered TPR of 0.54 (U2R) and 0.51 (Probe); removing the coordinator increased time-to-containment in all six configurations ($p < 0.01$).

CausalStress: Causal-Consistency Evaluation of Time-Series Forecasters

ETTh1, 5 seeds

- **Problem/Action:** Asked whether forecasting accuracy implies correct response to interventions; built a post-hoc framework (Granger dependency estimation, counterfactual stress generation) with new metrics, the Causal Consistency Score and Causal Reliance Index.
- **Result:** ARIMA had the best MAE (0.170) but ignored covariates, and all three neural models responded in the wrong direction on estimated drivers ($CCS \approx -0.04$); found and fixed a direction-reversal bug in my own intervention design.

Energy-Aware Adaptive Inference for Efficient Deep Learning

MNIST, CIFAR-10

- **Action/Result:** Designed a probabilistic dual-path CNN inference switch (no retraining) with a proven energy-reduction and error bound; cut a FLOPs-based energy proxy by 22.17% on MNIST at 98.66% accuracy (full CNN: 98.60%) and matched full-model accuracy on CIFAR-10 (3 seeds).

Coverage-Degradation Bounds for Conformal Prediction Under Energy-Aware Switching

NSL-KDD, UNSW-NB15

- **Problem/Action:** Proved a closed-form bound on how much a conformal coverage guarantee degrades when a

pipeline routes a fraction p_r of traffic to a cheaper model (coverage $\geq 1 - \alpha - p_r \cdot TV$).

- **Result:** The bound held in 86 of 91 tested settings, none significant after Bonferroni correction; corrected a naive 90.6% energy saving to a measured 74.8%.

TECHNICAL PROJECTS

CARTOGRAPH : Autonomous Cyber Defense

Mar 2026 – Aug 2026

Evolution of CyberShieldNet | Python | FastAPI | React/TypeScript | GraphSAGE/GAT | Docker

- Built a detection-to-response pipeline (temporal and graph correlation, explainable risk scoring, investigation, policy-gated response); evaluated in 11 experiments (bootstrap CIs, 5 seeds) on a cyber range and NSL-KDD.
- Showed Random Forest and the hybrid model were statistically indistinguishable on the cyber range (correcting my earlier claim), while the hybrid had the best F1 (0.802) of five models on NSL-KDD; temporal correlation alone cut alerts $5.2\times$, but graph-only correlation merged a window into one 281-node cluster.

QuantumShield: AI-Driven Post-Quantum Migration Platform

Sep 2025 – Feb 2026

Python | FastAPI | React/TypeScript | PostgreSQL | Qiskit | SHAP | Docker

- Built a platform that inventories cryptographic assets, scores quantum risk (explainable model vs. rule baseline), recommends ML-KEM/ML-DSA/SLH-DSA migrations, benchmarks 13 schemes, simulates the migration; 76 tests.
- Measured ML-KEM-512 key generation $\sim 2,000\times$ faster than RSA-2048 but SLH-DSA-128f signatures $\sim 244\times$ larger than ECDSA-P256; on synthetic data, logistic regression beat the rule baseline (macro-F1 0.790 vs. 0.723, $p = 0.0005$) while XGBoost did not ($p = 0.66$).

ResearchMind-Q: Evidence-Grounded AI Research Assistant

Feb 2025 – Aug 2025

Python | FastAPI | React/TypeScript | SQLAlchemy/Alembic | Docker sandbox

- Built a platform running literature $\rightarrow$ hypothesis $\rightarrow$ sandboxed experiment $\rightarrow$ statistics $\rightarrow$ conclusion, with claims tagged by evidence level, statistics never passing through an LLM, and two human approvals per experiment.
- Ran reproducible model-comparison, IDS-comparison, memory and critic ablation experiments; agents raise an error instead of fabricating when no LLM is configured; 87 tests.

CyberShieldNet: Multimodal Threat Intelligence and Risk Scoring

Jan 2024 – Jun 2024

Python | FastAPI | React

- Fused graph, temporal, behavioral and asset-context signals into a risk score (PyTorch, FastAPI, React); on a synthetic benchmark the fused model reached F1 0.765 vs. 0.575 for the best single modality, all leave-one-out ablations degraded, and a neural fusion engine raised F1 to 0.816.

Clinadapter-XRay: Efficient Chest X-Ray Report Generation

Jul 2024 – Dec 2024

- Built a dual-adapter framework (anatomy-aware visual adapter, LoRA-based ClinicalTermAdapter) training 7.45% of parameters in under 2 GB of GPU memory; clinical metrics converged earlier than linguistic (NLG) metrics.

xai-fraud-detection: Real-Time Explainable Fraud Detection Platform

Jun 2024 – Jan 2025

- Built a microservices platform (Flask, React, Kafka, Flink, Neo4j) combining Isolation Forest, Random Forest, an autoencoder, a GraphSAGE edge classifier and SHAP explanations, with a benchmarking module (bootstrap CIs).

WORK EXPERIENCE

Arventiq Labs — Founding Member & R&D Head

Bengaluru | January 2026 – Present

- Helped build the complete platform, now used by over 1000 users at more than 3 universities in Karnataka, with the goal of making quality education (UN SDG 4) affordable for students.
- Set the technical agenda for applied-research initiatives, requiring formal proof and rigorous empirical validation before any result was treated as reliable.
- Took manuscripts through peer-review revisions, correcting a reviewer's mathematical error and narrowing over-scoped claims to match what was proven.

TECHNICAL SKILLS

Languages: Python, Java, JavaScript/TypeScript, C/C++, SQL

ML & Research: LLM applications, RAG, scikit-learn, Explainable AI, Conformal Prediction, GNNs, Multi-Agent RL

Web & Backend: React, Next.js, Spring Boot, FastAPI, Node.js, REST APIs, PostgreSQL, MongoDB, Redis

Tools & Infrastructure: GCP, Docker, Jenkins (CI/CD), Git/GitHub, Linux/Bash, JUnit, Pytest, LaTeX

HONORS, AWARDS & CERTIFICATIONS

- **AMD AI Engage:** selected in the top 100 in India from about 100,000 participants.
- **Amazon ML Summer School 2026:** selected as one of about 2,000 students from roughly 350,000 registrants.